A Comprehensive Framework for Bangla Counterspeech: Data Creation, Model Training, and Evaluation

Abstract

Hate speech and harmful text are common on online platforms, including Bangla social media and online discussions. Counterspeech is a response that tries to reduce harm using normal and non-hostile language instead of removing content. However, there is no proper and publicly available dataset for Bangla counterspeech, which makes research in this area difficult.

This research focuses on creating a simple Bangla counterspeech dataset and training different machine learning models on it. The dataset will be written in JSON format and will contain the fields: id, text, counterspeech, and label. Models will be trained using this dataset, and their performance will be compared using automatic evaluation metrics. The final outcome will include an evaluation table and a discussion of results.

Introduction

Online platforms and social media are widely used for sharing opinions and ideas. Along with positive communication, these platforms also contain harmful, offensive, and hateful text. One way to deal with such content is counterspeech. Counterspeech means replying to harmful text using calm, polite, and logical language instead of deleting or attacking the speaker. The main purpose of counterspeech is to reduce harm and encourage positive conversation.

Counterspeech has been studied in recent years, mainly in the field of natural language processing. Most existing research focuses on high-resource languages such as English. Researchers have created datasets and trained different models to generate or evaluate counterspeech. Automatic evaluation methods like BLEU and ROUGE are often used to compare model outputs with reference counterspeech.

However, research on counterspeech in Bangla is very limited. Although Bangla is a widely spoken language, there is no standard and publicly available counterspeech dataset. Most Bangla studies focus on hate speech detection rather than counterspeech generation. Due to the lack of proper datasets, it is difficult to train models or compare their performance for Bangla counterspeech tasks.

This research aims to address this gap by creating a simple Bangla counterspeech dataset in JSON format. The study focuses on training different models using the same dataset and evaluating their performance using automatic metrics. The main contribution of this work is dataset creation and model comparison, which can support future research on Bangla counterspeech

Problem Statement

There is a lack of simple and structured datasets that can be directly used to train and evaluate Bangla counterspeech models. Most existing studies focus on English data, complex setups, or human surveys. Due to the absence of Bangla counterspeech datasets, researchers cannot properly train or compare models for this language. This research addresses this problem by creating a clear Bangla dataset and evaluating models only based on their training results.

Research Objectives

1. To create a Bangla counterspeech dataset in JSON format.
2. To train different machine learning or deep learning models using the Bangla dataset.
3. To evaluate and compare model performance using standard evaluation metrics.
4. To analyze the results and observe how different models perform on Bangla counterspeech data.

Research Questions

1. How can a Bangla counterspeech dataset be created in a simple JSON structure?
2. How do different models perform when trained on the same Bangla counterspeech dataset?
3. Which model gives better results based on evaluation metrics?

Research Methodology

Dataset Creation

The dataset will be created manually or collected from existing Bangla text sources. Each data entry will be stored in JSON format with the following fields:

- id: unique id
- text: original Bangla text
- counterspeech: Bangla counterspeech response
- label: category or class label

Model Training

Different models will be trained using the same Bangla dataset. These may include traditional machine learning models and basic deep learning models. The models will be trained and tested using standard data splitting methods.

Evaluation

The models will be evaluated using automatic text generation metrics including BLEU, ROUGE, BERTScore and many more. These metrics will be used to compare the generated Bangla counterspeech with reference counterspeech. The results will be presented in a comparative table to analyze model performance.

Expected Outcome

The expected outcome of this research includes:

- A Bangla counterspeech dataset stored in JSON format
- Trained models based on the Bangla dataset
- An evaluation metrics table comparing model performance
- A clear analysis of results

Work Plan

1. Bangla Dataset creation
2. Data preprocessing
3. Model training
4. Model evaluation
5. Result analysis and conclusion

Conclusion

This research presents a simple and clear approach to Bangla counterspeech modeling. Due to the lack of existing Bangla counterspeech datasets, this study focuses mainly on creating a structured dataset. Using this dataset, different models will be trained and evaluated to compare their performance. The research relies on automatic evaluation metrics and does not include human surveys or complex systems. Overall, this work aims to support future research on Bangla counterspeech and provides a practical framework suitable for thesis-level study.

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